

KAILASH AMBARAM CHAUDHARY

Cloud & DevOps Engineer | Python | AWS | RAG Pipelines | Docker & Kubernetes | CI/CD

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PROFESSIONAL SUMMARY

Early-career Cloud DevOps engineer with hands-on internship experience building CI/CD pipelines (GitHub Actions, Jenkins), containerising microservices with Docker, and deploying applications on AWS using Terraform. Implemented observability across EC2 fleets using CloudWatch dashboards, alarms, and SNS—monitoring CPU utilisation, network traffic, and error rates in real-time. Diagnosed IAM permission boundary errors, CodeDeploy lifecycle hook failures, and Docker networking issues through systematic log analysis. Deployed a production RAG system on Kubernetes with readiness/liveness probes, ConfigMaps, and Secrets. Holds AWS Solutions Architect Associate and IIT Roorkee Advanced Cloud Computing & DevOps certifications. Eager to contribute to cloud-scale reliability, automation, and observability within Cisco's SSE Control Plane engineering team.

TECHNICAL SKILLS

CI/CD	GitHub Actions, Jenkins, AWS CodeDeploy — automated build, test, deploy workflows; GitLab CI (familiar)
Containers	Docker (multi-stage builds, networking, Compose, Docker Hub), Kubernetes — pods, services, deployments, ConfigMaps, Secrets, health probes
Observability	AWS CloudWatch dashboards & alarms, SNS notifications, log-based troubleshooting; exposure to Prometheus/Grafana concepts
AWS / Cloud	EC2, S3, IAM, VPC, ALB, Auto Scaling Groups, CodeDeploy, CloudWatch, Security Hub, WAF, Boto3 SDK, AWS CLI
Networking	VPC, Security Groups, ALB, DNS, HTTP/HTTPS, TCP/IP, Load Balancing — practised through AWS infrastructure provisioning
IaC	Terraform — VPC, subnets, security groups, ALB, ASG, IAM roles, S3 artifact storage
Scripting	Python (primary), Bash/Shell scripting, YAML pipeline authoring, deployment automation
Backend / APIs	FastAPI, Flask, REST APIs, SQLAlchemy, MySQL, Gunicorn, pytest
Version Control	Git, GitHub — branching strategies, pull request workflows, structured commit history

WORK EXPERIENCE

Cloud & DevOps Intern · TechEasy Consulting · PuneOct 2025 – Dec 2025

- Engineered and maintained GitHub Actions CI/CD pipelines to automate build, test, and rolling deployment of a Java application to AWS EC2; resolved YAML misconfigurations and IAM permission boundary errors that were blocking the release pipeline.
- Provisioned and managed AWS infrastructure using Terraform and Boto3 (EC2, S3, IAM, ALB, Auto Scaling Groups), replacing manual console steps with version-controlled, repeatable infrastructure code.
- Built AWS CloudWatch dashboards to monitor real-time CPU utilisation, network traffic, and application error rates across EC2 fleets; configured alarms with SNS notifications for threshold breaches and authored RCA documentation and deployment runbooks adopted by the team.
- Configured AWS WAF rules to detect and block malicious/bot-generated traffic; defined IP-based rate limiting and geo-blocking policies that reduced fake request volume and improved CloudWatch log signal-to-noise ratio.
- Collaborated in an Agile workflow with development and QA teams, contributing to sprint goals around deployment reliability, CI/CD pipeline stability, and operational runbook quality.

Cloud Associate Engineer Intern · TechnoGrowth Software Solutions · PuneDec 2023 – Jan 2024

- Configured and maintained Docker containers and Linux-based server environments following team standards, supporting consistent and reproducible deployments across development and staging.
- Executed post-release deployment validation checks and authored onboarding documentation to help new team members ramp up on the existing infrastructure setup.

PROJECTS

Automated CI/CD Pipeline — Lift & Shift on AWS

Terraform · GitHub Actions · AWS EC2 · ALB · Auto Scaling Groups · S3 · CodeDeploy · IAM · CloudWatch github.com/kailashji24/test-repo-for-devops-app

- Provisioned complete AWS infrastructure as code using Terraform: VPC, subnets, security groups, ALB, Auto Scaling Group, IAM roles, and S3 artifact storage — zero manual console steps, fully repeatable.
- Automated the full software delivery lifecycle via GitHub Actions: Maven build → S3 artifact upload → CodeDeploy-triggered rolling deployment to EC2; diagnosed and fixed IAM boundary conflicts and lifecycle hook failures through CloudWatch log analysis.
- Configured CloudWatch dashboards to monitor deployment health, instance metrics, and ALB request counts; set alarms with SNS notifications for proactive incident response across the Auto Scaling Group.

LLM Document Q&A System — RAG Pipeline with Kubernetes Deployment

Python · FastAPI · LangChain · FAISS · HuggingFace · Groq Llama 3.1 · Docker · Kubernetes · GitHub Actions github.com/kailashji24/llm-document-qa-rag

- Built a production-style RAG microservice using FastAPI: PDF ingestion → semantic chunking → FAISS vector indexing → Groq/Llama 3.1 grounded answer generation; exposed via REST API with OpenAPI/Swagger documentation and health-check endpoints.
- Deployed to Kubernetes using Deployments, Services, ConfigMaps, and Secrets; configured readiness and liveness probes for reliable container lifecycle management — resolved FAISS index availability issues inside K8s pods.
- Built GitHub Actions CI pipeline for automated dependency validation, Docker image build, and deployment artifact verification; managed API credentials using K8s Secrets and environment-based configuration via ConfigMaps.

Flask Student Registration System — Jenkins CI/CD Pipeline

Python · Flask · MySQL · Docker · Docker Compose · Jenkins · Gunicorn github.com/kailashji24/flask-student-registration-devops

- Containerised a multi-service Flask REST API using Docker Compose (Flask + MySQL with persistent volumes, health checks, and service networking); deployed with Gunicorn as the production WSGI server.
- Built a Jenkins CI/CD pipeline that triggers on code push, builds and validates the Docker image, and redeploys the updated container end-to-end; resolved Docker networking errors and pipeline stage failures during setup.

Suspicious Activity Detection System

Python · TensorFlow · Keras · OpenCV · LSTM · CNN (LRCN) · Streamlit github.com/kailashji24/Suspicious-Activity-Detection

- Built an LRCN (CNN + LSTM) deep learning pipeline for human activity classification from surveillance video; deployed as an interactive Streamlit dashboard with confidence scoring and prediction analytics.
- Published research in IRJMETS (Vol. 07, Issue 02, Feb 2025): "Investigation on Human Activity Recognition using Deep Learning".

EDUCATION

B.E. in Computer Science Engineering — CGPA: 7.90 / 10

PREC, Savitribai Phule Pune University, Loni, Maharashtra | 2025

CERTIFICATIONS

- AWS Solutions Architect Associate | Intellipaat | March 2025
- Advanced Certification in Cloud Computing and DevOps | iHUB DivyaSampark, IIT Roorkee | July 2025
- AWS Solutions Architecture Job Simulation | Forage | July 2025

ADDITIONAL

Actively practising Data Structures & Algorithms in Python on LeetCode. All projects hosted on GitHub with structured READMEs and clean commit history. Available immediately; open to relocation to Bengaluru.